

Overview of Changes Made to NB1

Vectorization by chunks, a way to do array-wise computation instead of pixel-wise computation that takes into account the computer's memory and computing power, was implemented for each function in ClimateRegime.py that iterated through elevation data. The exact functions updated to include vectorization by chunks in Climate Regime are:

1. setClimateAndSoilWaterData
2. getThermalClimate
3. getThermalZone
4. getTemperatureProfile
5. getLGP
6. TZoneFallowRequirement
7. AirFrostIndexandPermafrostEvaluation
8. AEZClassification
9. getMultiCroppingZones
10. getAnnualTemperatureAmplitude
11. getNPP
12. getBeginningDateofHibernationPerio
13. getHibernationPeriodLength

Updated in UtilitesCalc.py:

1. averageDailyToMonthly
2. Added Function: compute_chunk_size_multi

Updated in ETOCalc.py:

1. calculateETONumba
2. Added Function: compute_Rn_chunk

Updated in LGPCalc.py:

1. Added Function: process_chunk

The original script had bottlenecks and optimization opportunities that needed to be addressed. Some of these bottlenecks resulted from heavily nested loops for pixel computations that involved many conditional branches, which can eat up computing time. Not only that, but with each pixel having to go through each set of conditionals for

classification functions such as `getMultiCroppingZones` and `getAEZClassification`, it can cause major slowdowns for countries with larger area boundaries. To go with these changes in the main `ClimateRegime.py` script, a function was added to compute the optimized chunk size for each adapted function, which can be found in `UtilitiesCalc.py` named `compute_chunk_size_multi` to find the best size of array that still provides a speed up for the function and to balance the memory needed for overhead. Another optimization implemented in this new code was adding handler functions for both `ETOCalc` and `LGPCalc`, which are scripts composed of Numba functions, to better support this new strategy of arraywise computation. Numba handles fast computations by compiling rather than interpreting, but it also struggles with vectorization, so helper functions were created to enable parallelization. To also support this need, a method of array-wise computation for the main `ClimateRegime` functions was the adaptation of the `averageDailyToMonthly` function to handle varying daily vector lengths when computing averages.